Closing the Dispersion Gap in Neural Limit-Order-Book Simulators: Findings from the August 2026 Fano Investigation

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Abstract

We investigate why neural marked-temporal-point-process (MTPP) simulators of Coinbase limit-order-book flow under-disperse relative to real markets, using the Fano factor profile $F(\Delta) = \mathrm{Var}(N_\Delta)/\mathbb{E}[N_\Delta]$ across bucket scales $\Delta \in [1, 50]$ s as the headline statistic. Across ten experimental arms on three assets we establish: (i) the bounded-rate SS2P2 head is not reliably self-exciting — its measured impulse response is sign-inconsistent and saturated at the intensity ceiling; (ii) one-step maximum likelihood systematically misallocates excitation spectra (three independent demonstrations); (iii) extending the TBPTT gradient horizon to cover the measured scales transforms the simulated dispersion profile (all BTC seeds rise ×15–32, best within 1.6× of real at 50 s); (iv) at equal training budget, dispersion peaks *early* and prediction accuracy peaks *late* along a single MLE trajectory — the prediction/simulation tension exists within one training run; and (v) the remaining level offset is dominated by ceiling saturation (a third of events occur at the cap; teacher-forced dispersion is only 1.2–1.4× low). We relate every finding to the calibration methodology of Jain et al. (Compound Hawkes), whose binned-count regression targets exactly the cross-scale count covariances that MLE ignores, and derive a ranked roadmap toward a dispersion-faithful, certified world model for market making.

1 Framing: two identities

All arguments in this note reduce to two identities. First, the *Cox decomposition* (law of total variance given the intensity path $\lambda(t)$, with $\Lambda_\Delta = \int_{\text{bucket}} \lambda\, dt$):

$$F(\Delta) \;=\; 1 + \frac{\mathrm{Var}(\Lambda_\Delta)}{\mathbb{E}[\Lambda_\Delta]}, \tag{1}$$

so all excess dispersion is variance of the integrated intensity; a post-hoc rate rescaling $\lambda \to \kappa\lambda$ scales $F - 1$ linearly and cannot repair relative fluctuation structure. Second, the *branching form* for a linear Hawkes process with branching ratio n: cluster sizes S satisfy $\mathbb{E}[S] = 1/(1-n)$ and $\mathbb{E}[S^2] = 1/(1-n)^3$, hence

$$F(\infty) \;=\; \frac{\mathbb{E}[S^2]}{\mathbb{E}[S]} \;=\; \frac{1}{(1-n)^2}. \tag{2}$$

The profile's *rise* across Δ is the fingerprint of long-memory (power-law-like) intensity autocovariance; single-exponential kernels plateau.

Real Coinbase flow (7 days, Jan 2026): $F(1\text{s}) \approx$ 40–70 rising to $F(50\text{s}) \approx 370$ (SOL), 1250 (ETH), 1460 (BTC), still rising at the window edge. Inverting (2) gives implied branching (lower bounds): SOL ≈ 0.948, ETH ≈ 0.971, BTC ≈ 0.973 — the $1/(1-n)^2$ amplification makes this small spread a 4× dispersion difference.

2 Data and the tick-regime contrast

	trades (7d)	book rows	rows/trade	rate (ev/s)	IS share
BTC	41,675	52.9M	1268:1	40.6	52%
ETH	94,048	55.5M	590:1	42.8	34%
SOL	40,048	20.1M	501:1	22.3	7%

Table 1: Raw Coinbase spot data, 2026-01-01..07, and the event-type composition of the processed 62-channel streams. BTC/ETH are small-tick assets dominated by inside-spread (IS) quote wars — the most self-exciting flow; SOL is effectively large-tick (1-tick spread, queue churn: 43% cancels). The tick regime, not just the rate, drives the criticality ordering.

3 Mechanism: the bounded head is not reliably self-exciting

SS2P2 factorizes intensity into a softmin-bounded scalar rate $\lambda(t) = s \cdot \mathrm{softplus}\big(c - \mathrm{softplus}(c - w^\top h(u))\big) \le s \cdot \mathrm{softplus}(c)$ and rate-neutral softmax marks. The stability certificate bounds the rate *level*; nothing constrains the event→intensity feedback. Measuring the feedback directly (inject one extra event into real histories; integrate the extra intensity mass over 20 s) across six trained checkpoints: median response ≈ 0 *to negative* $(-8.7 \ldots + 0.2)$, positive in only 27–56% of states, sign flipping across seeds; post-event intensity pinned at the ceiling in every window. The model's residual dispersion comes from a stereotyped impulse–decay envelope, not compounding cascades. In Hawkes terms: *a settable branching ratio is a property of linear-in-history intensity; for a bounded neural head it is not even well defined.*

4 Experimental arms

Protocol throughout (the `ma_cbse` benchmark): 62-channel events, TBPTT, categorical marks, streaming genuine evaluation on every test event, validation- calibrated 600 s rollouts, 3 training × 3 rollout seeds per asset.

4.1 Negative and structural results

- **Book conditioning is not the lever** (`ss2p2-lob`): six top-10-ladder features left prediction unchanged (BTC 0.4484 vs 0.4478) and made closed-loop volatility structure *worse* (F6 0.16 vs 0.28) — extra state conditioning becomes a smoothing feedback path in rollout.

- **lgm** (linear Hawkes ground × SS2P2 marks on the shared backbone; $n = \sum_m a_m/\beta_m$ exact and projected; pin $\mu_0 = R(1-n)$): the pin holds in free rollout ($\kappa = 1.01$ vs SS2P2's 0.52); marks near parity; Fano *shape* right with seed spread ±2 (vs SS2P2 ±217) but level low. Two porting lessons: a log-domain clamp silently capped the burst accumulators (excitation useless to MLE until fixed); and **zero cold-starts of the ground are catastrophically biased** — the stationary mean is $\mathbb{E}[S^m] = R/\beta_m$ (≈ 600 for slow kernels), so windowed validation punished slow-memory solutions and froze best-model selection at epoch 1. Fix: stationary-mean cold start.

- **MLE misallocates spectra** (three demonstrations): the fixed LGM pushed kernel mass to its slowest decay; the typed-kick M=6, n=0.97 arm (`lgm-k`) went slower still ($\beta \approx 0.008$,

∼2-min memory), *lowering* observed-scale Fano despite the higher ceiling; the dispersion-loss arm (below) pinned only the scales in its grid. One-step likelihood is structurally blind to $F(\Delta)$ and, given capacity, prefers slow smooth intensity.

- **Teacher-forced variance matching ≠ closed-loop compounding** (`ss2p2-disp`): an auxiliary loss matching model-implied $1 + \mathrm{Var}(\Lambda_\Delta)/\mathbb{E}[\Lambda_\Delta]$ to the batch-empirical Fano at $\{0.5, 2, 8\}$ s pinned those scales reproducibly (seed sd ±14 vs ±217) at zero prediction cost — and went flat beyond its grid.

4.2 The window-coverage result

At seq 1024 the TBPTT gradient truncates at ∼24–25 s on BTC/ETH — before the 20–50 s scales where their simulated profiles flatten; SOL's ∼46 s windows cover the range (and produced real-like rises in 2/3 seeds). Extending to seq 4096 (∼100 s windows; recipe otherwise identical):

BTC, per seed	seq 1024 (12 ep)	seq 4096 (12 ep)	real
$F(50s)$, s1/s2/s3	29 / 50 / 361	**901 / 270 / 744**	1463
rise $F(50)/F(1)$	×2.5–13.9	×15–32	×20.9

Table 2: Window coverage transforms the profile: all three BTC seeds rise steeply (best within 1.6× of real). ETH: 2/3 seeds transformed. SOL: unchanged — the clean negative control. The BTC "seed lottery" was gradient starvation at unobserved scales.

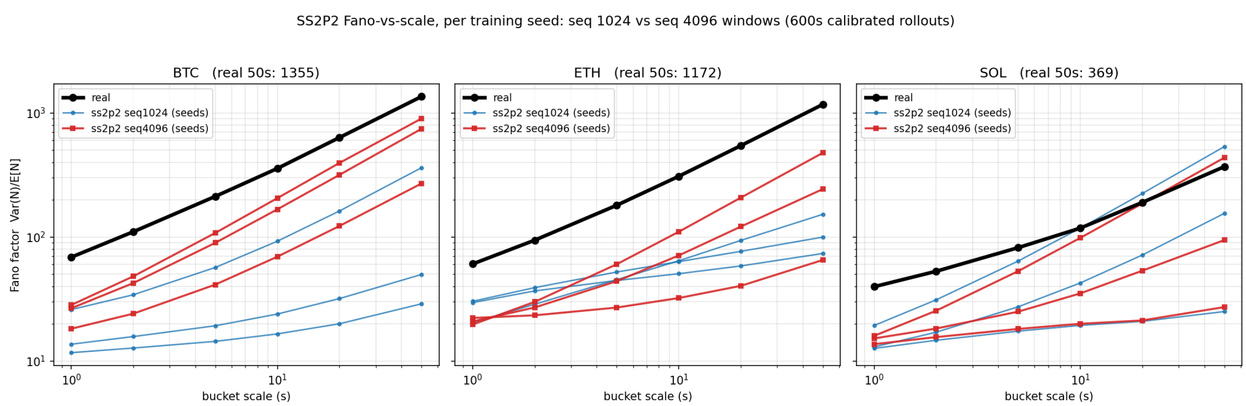

Figure 1: Per-seed Fano-vs-scale (log–log): black = real, blue = seq-1024 baseline, red = seq-4096. BTC: the whole seed family moves; SOL: negative control.

4.3 Equal-budget controls: the training-path trade-off

Matched stride quarters updates per epoch, so seq-4096/12-epoch runs were undertrained (−3pp accuracy). Controls at 48 epochs (exact update parity with the baseline):

Finding (the sharpest form of the arc's core lesson). Along a single MLE trajectory, *dispersion peaks early and accuracy peaks late*; no NLL-selected checkpoint has both. The prediction/simulation tension is a property of the training path, not only of architectures. The 12-epoch w4k checkpoints constitute a de-facto dispersion-optimal early stop; a paper can honestly present the frontier.

BTC arm	acc	$F(1\mathrm{s})$	$F(50\mathrm{s})$ per seed
base 1024 / 12 ep	0.4478	17.1	29, 50, 361
w4k 4096 / 12 ep	0.4183	24.3	**901, 270, 744**
w4k48 4096 / 48 ep	**0.4514**	16.6	232, 87, 230
w4kd + disp loss / 48 ep	**0.4519**	11.1	52, 43, 43
real	—	70.0	1463

Table 3: Equal-updates controls. Accuracy fully recovers and exceeds baseline; the extra training walks the model partly away from the burst-faithful solution. The dispersion loss under cap 6 suppresses long-scale Fano, as predicted by the saturation diagnostic.

4.4 Where the residual level offset lives

Diagnostics on the w4k BTC checkpoint: (i) **ceiling saturation** — dominating rate 359 ev/s; event-time intensity $p_{90} = 357$, $p_{99} = 358$; **32.8% of events above** $0.9\times$**ceiling**: bursts are clipped and gradients (including any dispersion loss) are dead there; (ii) **teacher-forced dispersion** is only 1.2–1.4$\times$ below real at every scale (e.g. 50 s: 240 vs 290). Conditioned on real history the model carries ~75% of real dispersion; the cap, then modest closed-loop drift, accounts for the rest. The cap-9 control (`ss2p2-w4kc9`, in flight) tests whether headroom changes the MLE optimum itself.

5 The Jain (Compound Hawkes) reference point

Jain's simulator never optimizes or reports the Fano factor; dispersion is scored via volatility signature plots and $|r|$-autocorrelation, and his own signature plots undershoot empirical levels by 2–3$\times$ — the published state of the art also misses level while matching shape. What his pipeline gets structurally right:

1. **Estimation by binned-count regression** (Kirchner-style, lin–log lag grid): binning turns the Hawkes process into a vector autoregression on lagged counts whose coefficients are the kernel integrals; the fitting target *is* the cross-scale count autocovariance — the integrand of $F(\Delta)$. Moments, not MLE; the spectrum cannot be misallocated relative to dispersion.

2. Power-law kernels selected by AIC 100% of the time (straight log–log point estimates over six decades of lag, 10^{-4}–10^{2} s).

3. Kernel-norm matrix eigenvalue capped at 1 by rescaling (his projection); inhibitory cross-kernels retained.

4. Baselines derived, not fitted: $\mu = (\mathbb{I} - M)\Lambda$ — the matrix form of LGM's pin.

5. Compound order sizes (Dirac spikes at round numbers + geometric); the constant-size ablation is wrong by ~100$\times$ on price variance. Time-of-day multipliers throughout.

The RL chapter's acceptance checklist. His market-making chapter converts stylized facts into functional requirements, ranked by agent ablations: (1) *clustered arrivals as the agent's observable* — removing Hawkes intensities from the RL state flips Sharpe $+31.5 \to -20/-51$, the most catastrophic ablation; (2) *queue dynamics* — removing relative queue position yields pump-and-dump or unprofitable policies; (3) *arbitrage-consistent impact memory* — exponential kernels admit dynamic arbitrage that the optimizer finds ("pump & dump"), rarely so under power-law: mis-specified memory manufactures fake alpha; (4) *responsiveness* — spread widening, queue depletion, liquidity regeneration in reaction to the agent; (5) spread mean-reversion via spread-dependent in-spread intensities ($\propto (s-1)^{0.75}$, $R^2 = 0.85$); (6) realistic frictions (strategies collapse beyond 2 bps). Items (3)–(4) require genuine self-excitation — absent in our measured SS2P2 checkpoints — making the dispersion program a prerequisite for the agent phase, not an aesthetic.

6 Conclusions

1. Prediction and simulation are different objectives; one-step MLE is blind to, and given capacity misallocates, the spectrum controlling $F(\Delta)$.
2. The TBPTT gradient horizon must cover the measured scales: the cheapest large win found (BTC 3/3 seeds transformed).
3. Along one training run, dispersion peaks early and accuracy late; no NLL-selected checkpoint dominates.
4. The rate ceiling is the current level bottleneck (a third of events saturated; teacher-forced dispersion nearly right).
5. Certified dispersion requires linear-in-history structure: only LGM has a settable n, a pinned mean rate, and closed-form $F(\Delta)$ at any horizon; the Kirchner regression is the proven recipe for fitting its ground (typed, per asset) to cross-scale covariances post hoc, marks untouched.
6. Per-asset targets are mandatory (n: 0.948 vs 0.973 is a $4\times$ Fano difference); the tick regime (IS share) should shape typed-kick priors.

Roadmap. (1) cap-9 control (in flight); (2) Kirchner binned-regression fit of the LGM ground; (3) LGM ground $\times$ bounded neural gate ($\rho_{\text{eff}} \leq n\, G_{\max}$) to restore timing without breaking the certificate; (4) time-of-day multiplier; (5) exploitability audit before trusting any policy trained in the simulator.

Provenance: experiments under `peacock:~/simulation/experiments/ma_cbse/`; code `honglinfu98/simulation` commits `7b97b21..88d22cc`; full lab log in `docs/fano_investigation_2026-08.md` of this repository.